

Portfolio Optimization and Market Indexes Tracking

Computer Science Final Year Project

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Motivation

- Asset values may decline abruptly and without notice, as shown by the COVID-19 pandemic
- The world has changed in the past year, affecting our lives, economy, and businesses
- Diversifying or optimizing the portfolio is crucial, but there is little consensus on how to do it
- Most investment portfolios are suboptimal and can be improved
- Interest in MATLAB and portfolio optimization led to choosing it as final year project
- Got more inputs from research works provided by Professor Wang

Problem Statement & Scope

- Markowitz optimization is empirically unstable and sensitive to small changes in input parameters
- Markowitz optimization is a weakly posed inverse problem that needs regularization strategies to control instabilities
- Use of the new formulations by Prof. Wang and Team
- Project will apply several risk measuring criteria, such as Calmar ratio.
- Project will combine Calmar Ratio formula with a weighted regularization parameter to balance Calmar ratio and portfolio return maximization
- Project will compare different portfolios based on these risk measures for various markets, such as S&P 500, FTSE100, HSCI, and HDAX
- Project will use neurodynamic optimization (NO) portfolio and compare it with other portfolios, such as Equally distributed weighted optimization and eight other baselines

Methodology

Example: Pain Ratio for FTSE100

- Portfolio optimization with Pain Ratio
- Nonlinear optimization with NNN_IG
- Data split into training and test sets
- Optimal portfolio weights and ksi for each week
- Performance evaluation and comparison with other portfolios
- Output files and arrays for portfolio weights, Pain Ratios, cumulative returns, and nonlinearity
- Graphs to visualize the results
- Different factors for Pain Ratio and expected return depending on dataset

Formulation

Converting the Formula to Code blocks

To run the Convex Optimization these datasets we need the following (using `fmincon`):

$$x = \text{fmincon}(\text{fun}, x0, A, b, Aeq, beq, lb, ub, \text{nonlcon}, \text{options})$$

Where,

fun is the nonlinear objective function to be minimized.

x0 is the initial point for the algorithm.

A is the linear inequality constraint matrix.

b is the linear inequality vector.

Aeq is the linear equality constraint matrix. (Shown in Demo)

beq is the linear equality vector. (Shown in Demo)

lb is the lower bound for the variables.

ub is the upper bound for the variables.

nonlcon is a function that returns the nonlinear inequality and equality constraints, respectively.

options is a structure that specifies additional options for the algorithm, such as 'sqp' for the Algorithm option.

Formulation

Converting the Formula to Code blocks

- Formula Description:

$$\max_{y, \eta, \xi} \mu^T y - r_f \eta \quad \leftarrow \text{Objective Function}$$

$$\text{s.t. } e_n^T y = \eta, y \geq 0, \quad \leftarrow \text{Inequalities}$$

$$\text{Non-linear Function} \longrightarrow \frac{1}{m} \sum_{j=1}^m \left(\xi_j - \sum_{i=1}^j r_i^T y \right)^+ \leq 1,$$

$$\xi_{j-1} \leq \xi_j, j = 1, 2, \dots, m;$$

Pain Ratio Formula

Formulation

Converting the Formula to Code blocks

- Objective Function:
$$\max_{y, \eta, \xi} \mu^T y - r_f \eta \quad \leftarrow \text{Objective Function}$$
 - Code: `objective = -(mean_return'*y-risk_free_rate(week)*ita);`
- Non-linear Function:
$$\frac{1}{m} \sum_{j=1}^m \left(\xi_j - \sum_{i=1}^j r_i^T y \right)^+ \leq 1, \quad \leftarrow \text{Non-linear Objective}$$
 - Code:
`con_sum(j) = max(ksi(j)-sum(rj'*y),0);`
`c = sum(con_sum)/(week-1)-1;`

Evaluation and Results

Datasets

Library	Dataset	No. of Stocks	No. of Weeks	Time Span	Reference
BCST-Library	S&P500	442	595	01/11/2004 - 29/04/2016	(Bruni et al., 2016)
	NASDAQ100	82	596	01/11/2004 - 29/04/2016	
	FTSE100	83	717	01/07/2002 - 29/04/2016	
	DJIA	28	1363	01/02/1990 - 29/04/2016	
Updated Library	HDAX	49	938	03/01/2000 - 29/11/2017	(Leung et al., 2022a)
	FTSE100	56	938	03/01/2000 - 29/11/2017	
	HSCI	77	938	03/01/2000 - 29/11/2017	
	S&P500	356	938	03/01/2000 - 29/11/2017	

Evaluation and Results

Optimized Values and Results

Ratios	Dataset	Portfolios		
		NO	EW	MI
Calmar Ratio	HDAX	0.0766	<u>0.0649</u>	0.0549
	FTSE100	0.1029	<u>0.0927</u>	0.0427
	HSCI	0.0978	<u>0.0709</u>	0.0314
	S&P500	<u>0.0827</u>	0.0994	0.0691
Martin Ratio	HDAX	0.2721	<u>0.2408</u>	0.2073
	FTSE100	0.3706	<u>0.3441</u>	0.1497
	HSCI	<u>0.1873</u>	0.2278	0.0936
	S&P500	0.0255	0.0187	<u>0.0232</u>
Pain Ratio	HDAX	0.4288	<u>0.3764</u>	0.3240
	FTSE100	0.5585	0.5189	0.2228
	HSCI	<u>0.2683</u>	0.3335	0.1264
	S&P500	<u>1.384</u>	1.426	1.298
Conditional Pain Ratio	HDAX	0.0975	<u>0.0865</u>	0.0793
	FTSE100	0.1311	0.0985	<u>0.1196</u>
	HSCI	<u>0.0855</u>	0.0924	0.0783

Ratios	Dataset	Portfolios								
		NO	EW	MI	M-V	L-SSD	LR- ASSD	RMZ_S SD	PK_SSD	CZeSD
Calmar Ratio	DJIA	0.0154	<u>0.0248</u>	0.0132	0.01722	0.0134	0.0195	0.0088	0.0284	0.0215
	NASDAQ 100	0.1939	0.1517	0.1391	0.1190	<u>0.1542</u>	0.1269	0.1075	0.1077	0.1260
	FTSE100	0.0843	0.01230	0.0434	0.2104	0.1660	0.0440	<u>0.1918</u>	0.1904	0.0918
	S&P500	0.1297	0.0986	0.0910	0.0947	0.1017	0.0968	0.1408	0.0784	<u>0.1300</u>
Martin Ratio	DJIA	<u>1.3900</u>	1.4373	0.4183	0.7935	0.3953	0.8723	0.1310	0.5979	1.2003
	NASDAQ 100	<u>1.3894</u>	1.5476	1.1642	1.3120	1.8891	1.0654	0.8113	0.4647	1.1492
	FTSE100	1.1374	0.8420	0.2126	2.7423	1.6577	0.2740	<u>2.2513</u>	0.7786	0.5102
	S&P500	<u>0.0947</u>	0.0901	0.0933	0.0832	0.0801	0.0730	0.0897	0.0301	0.1185
Pain Ratio	DJIA	<u>0.2934</u>	0.3078	0.1135	0.1567	0.1063	0.2017	0.0375	0.1669	0.2309
	NASDAQ 100	<u>1.1956</u>	1.1576	0.8020	0.8246	1.2699	0.8440	0.8075	0.7979	0.9103
	FTSE100	0.7921	0.6494	0.1991	<u>1.2198</u>	1.0319	0.4166	1.2473	0.8973	0.4614
	S&P500	<u>0.7907</u>	0.7572	0.6733	0.5331	0.5016	0.5733	0.6292	0.5023	0.9491

Conclusion

- Learnt more about MATLAB, applications in academic and financial world
- Tested and verified code, implemented Calmar, Pain, Martin, and Conditional Pain ratios, NO portfolios superior to baselines in majority of cases
- NO portfolios outperformed baselines in most cases
- Updated Library - 15 scenarios - 10 best result and 5 second-best result
- BCST Library - 12 scenarios - 7 best & second-best results